

MEASURE — can we believe a number

Owns: `engine/mew.js`, `engine/sprt.js`, `engine/provenance.js`, `engine/status.js`, `engine/backtest_winrate.js`, `engine/paired_h2h.js`, the noise floor, the corpus stamps, and the MAG refit.

Its one number: leaf calibration — when the leaf says 90%, is it 90%.

May not: change a policy, a search knob or an engine mechanic. This division builds the rulers; it does not compete on them.

<!-- GENERATED: engine/status.js -->

```
MEASURE — can we believe a number
leaf calibration: live in-game leaf is WORSE than a coin on Brier (paired +0.0502, 95% CI 0.0371 to 0.0633)
n=1378 games, 200 rollouts each (2026-08-04 07:09)
when it says 90-100% it wins 54% (n=56); when it says 0-10% it wins 54% (n=52) — ECE 0.1811
powered for MDE 53.8% held-out / 51.7% full corpus; the prior effect needed n=2835
PRE-CHANGE — measured against a different build of: engine/medicham2-browser.js, engine/rollout_leaf.js
(the corpus has grown since: data/games.ladder.jsonl — more power available, not staleness)
provenance: 2 unsafe, 42 possibly stale, 50 ok, 0 missing
refit edge: CLEAN — feature_fixture --check passes: all 58 columns hash-identical to fit time
(engine/medicham2-browser.js moved 2026-08-04 23:41, but the feature function did not)
```

stamped 2026-08-05 00:26

<!-- /GENERATED -->

Why the refit lives here, not in ENGINE or SEARCH

The refit is the expensive event on the one expensive edge, and it invalidates seven artifacts: counterplay, winrate-backtest, opponent-calibration, weight-multiplicity, then the mag / mew / scoreboard bundles.

`provenance.js` derives that set rather than carrying a typed list of it.

The division that owns *knowing when a number stopped being true* is the one that should be pulling that trigger.

A restamp is only valid if the feature FUNCTION is unchanged. Damage table moved → refit. Not a restamp. There is no version of this where the shortcut is fine.

Open — in priority order

1. LEAF CALIBRATION — MEASURED 2026-08-04. The leaf is not calibrated, and the claim is now powered.

`data/winrate-backtest.json` was re-derived against the current engine, on the whole clean corpus instead of a 350-game subsample, and it publishes a reliability curve. The finding is worse than the verdict string it replaces, and worse in a specific and actionable way.

The old number scored a leaf no live decision calls. It measured `winProb2` — `battle()` at MEDICHAM's default 20-turn horizon with entry effects re-fired. MILTANK calls neither: its team-preview leaf is a greedy payout at `maxTurns=60` with `seeded:true`, and its in-game leaf is

`rollout_leaf.rolloutWinProb` at `explore=1.0 / foePolicy=uniform / maxTurns=60`. All three are now scored on identical positions, so the difference between them is about the leaf.

Confidence carries no information. The curve is close to a horizontal line:

leaf	says 0-10%	says 90-100%	discrimination	Brier vs coin (paired)
in-game, 200 rollouts, held-out n=1,378	wins 53.8% (n=52)	wins 53.6% (n=56)	50.99% [48.3, 53.7]	+0.0502 [0.0371, 0.0628]
preview, 40 rollouts, full clean n=6,886	wins 45.7% (n=831)	wins 55.3% (n=933)	53.22% [52.0, 54.4]	+0.0740 [0.0668, 0.0813]

Positive is worse. Both leaves are decisively **worse than a coin** on Brier and on log-loss, and worse than player-Elo, paired on the games where both have an opinion. The preview leaf puts **25.6% of all its predictions into the two extreme buckets**, where it is wrong by ~40 points.

Discrimination and calibration are separate failures needing separate answers. The preview leaf does rank — 53.22% on 6,700 decisive calls, $p < 1e-4$ — real, but only ~1.9 points above the split-half noise floor. The in-game leaf does not rank at all: 50.99%, $p = 0.47$. Randomising the playout bought variance and spent the signal.

What this is not. Nothing here says the engine is broken. The legacy `winProb2` leaf reproduces the 2026-08-02 number closely on the current engine — held-out log-loss **1.0243** against the **1.0748** published, discrimination **51.94%** against **52.63%** — so the twenty-two engine commits in between did not move the headline. Do not spend this finding on a mechanics hunt.

Also fixed here: the split was cutting on **store append order**, not date, and the store carries 4,775 date inversions. A side-symmetry witness scores 400 boards from both sides and reports $\text{mean}(p_1+p_2-1) = -0.0099$, so no side advantage inside the engine is contaminating the result.

Still open, in order:

- the leaf is scored at **turn 0**, because that is where a game-outcome label exists. `rollout_r1.js` scores mid-game positions and is the other half of this; the two have never been read together.
- the **horizon** is the first suspect. `battleResult` falls back to bodies-then-HP whenever the playout does not finish, so a confident number can be a material count wearing a probability's clothes. That is a SEARCH change, not a MEASURE one — file it, do not fix it here.
- `data/winrate-backtest-rows.jsonl` holds the per-game predictions, so the curve can be re-cut without re-running the ~15 minutes of rollouts.

2. R4 has an artifact — CLOSED 2026-08-04

`engine/rollout_r4.js` writes `data/rollout-r4.json`, and `status.js` now prints the verdict out of that file instead of `NO ARTIFACT`. It does not re-pair anything; it shells out to `sprt.js --verify` and `paired_h2h.js` and refuses to write if they disagree, the same way `status.js` shells out to `provenance.js`.

The remembered 55.5% held. **ACCEPT H1 — MILTANK takes 55.5% of 535 DECISIVE PAIRS, decided after 522 of them, LLR 3.00 against a 2.94 bound.** The corpus is 5,248 lines, which is 2,624 games, which is 1,312 seed pairs — the store writes a log-only companion record under the same id, so a line count double-counts every game and the handoff's "5,248 games" was exactly twice the truth. The artifact records all four numbers and asserts the invariant that makes them relate.

Two things it is not. The point estimate is **stopped at a boundary**, so it is biased high and the 95% CI beside it is a fixed-n formula quoted for context, not the inference — read the verdict. And `status.js` classes the corpus **PRE-CHANGE**: `engine/medicham2-browser.js` moved 04:47, the games were played 04:41. Both arms shared the pre-fix rollout model, so the contrast is fair and the run stands as a measurement *of that build*; that the edge survives into HEAD is an assumption. It gets re-run at the next frozen engine release.

No A/A run exists for this comparison, so the noise floor is **not established**. The substitute in the artifact is three independent split-half cuts of this run: spreads of 0.2, 3.9 and 1.3 points against an effect of 5.5. One cut alone would have been useless — the spread of a single split-half is itself a draw with sd about 4.3 points at this sample size.

3. R1 has an artifact, and it does not say what the docs said — CLOSED 2026-08-04

`engine/rollout_r1_artifact.js` writes `data/rollout-r1.json` from the committed row dump, and `status.js` prints its verdict. The gate previously read a file of the same name that `engine/rollout_r1_join.py` wrote for the **withdrawn** cross-language join — nothing was hidden, the join prints its own withdrawal, but the gate read it because it owned the filename. The join is now `data/rollout-r1-withdrawn-join.json` with `withdrawn: true`, and `status.js` refuses to print any artifact carrying that field.

The recomputation does not reproduce the published PASS. `docs/ROLLOUT-design.md` claimed 68.18% against material's 65.26%, +2.91 [1.79, 4.04]. From `data/rollout-r1-rows.jsonl` the same formulas give **65.72% against 65.26%, +0.46, 95% CI [-0.72, +1.63] — UNDECIDED** on 9,201 positions.

The material column matches the published figure to the digit, so it is the same sample. The rollout column reproduces §4.2.1's **greedy** calibration table bin-for-bin, so the surviving dump is the `explore=0` incumbent and the `explore=1` run that produced 68.18% left no file. That is the lesson, not the arithmetic: **the dump stamped no N, no explore and no build digest, so two runs four accuracy points apart were byte-indistinguishable.** `rollout_r1.js` now writes `data/rollout-r1-rows.meta.json` beside every dump. R2 and R3 still have the same hole.

The split-half spread of this run ranges 0.43 to 2.01 points against an effect of 0.46 — the effect is inside its own noise floor, which is an independent route to the same UNDECIDED.

Open consequence, filed to SEARCH: `--rollout-explore` defaults to 1.0 and `engine/rollout_leaf.js:147`, `engine/mag_bot.js:145` and `docs/MILTANK.md` all cite 68.18% as the reason. Re-running `EXPLORE_LIST=1 DUMP=rollout-r1-rows.jsonl node engine/rollout_r1.js` and then `node engine/rollout_r1_artifact.js` settles it, and R2 says the leaf is cheap.

SEARCH RAN IT, 2026-08-04. The published figure reproduces. `data/rollout-r1-explore-sweep.json` and `data/rollout-r1-explore1.json` : on the identical 9,201 positions, `explore=1.0` judges at **67.971%** against the published 68.18%, and its lift over the same material baseline is **+2.706 [1.596, 3.817]** against the published +2.91. Paired against the greedy arm it is **+2.25, 95% CI [1.31, 3.19]**, monotone in `explore` ($0 \rightarrow 0.5 \rightarrow 1.0 = 65.72 \rightarrow 67.58 \rightarrow 67.97$) and it holds at the live 60-turn horizon. **The retraction was right about the provenance and wrong as a guide to the arm — R1 is UNDECIDED on the incumbent and a PASS on the arm that ships. Three things this division should act on:** 1. **The command in this section would have destroyed the evidence.** `DUMP` resolves under `data/`, so `DUMP=rollout-r1-rows.jsonl` overwrites the committed greedy dump — the only record of the incumbent arm, committed for exactly that reason. SEARCH used a new filename. Worse, the sidecar path in `rollout_r1.js:338` is the hardcoded literal `data/rollout-r1-rows.meta.json` whatever `DUMP` is set to, so it lands beside the wrong dump. `rollout_r1_artifact.js` rejects it on the name-and-row-count check, which is the check working — but the fix is to derive the sidecar path from `DUMP`. 2. `status.js:229` **still prints the greedy arm as "R1 leaf accuracy"**. SEARCH did not overwrite `data/rollout-r1.json`, deliberately: it is this division's artifact, written hours earlier, and it is the only record of the incumbent. But the line now reads UNDECIDED for a configuration the bot does not run. One line — point it at `rollout-r1-explore-sweep.json`, or print both arms. 3. `engine/mew.js` **exposes no** `--miltank-explore`. So the question this settles — which playout JUDGES better — cannot be escalated to the one that matters, which playout WINS more. R4 was itself run at `explore=1.0` and cannot arbitrate its own setting. Two parsed flags on `mew.js` and the A/B becomes runnable. One hypothesis this division filed to

SEARCH is **measured and rejected**: `battleResult` scoring bodies-then-HP on unfinished playouts is real but is not the mechanism. Over 1.1M playouts, 99.5–99.8% end by an actual wipeout at every explore setting and at horizons 20 and 60; cap-hits are 0.2–0.5%. Exploration makes playouts longer (4.4 → 6.1 mean turns), not truncated. Filed to ENGINE as a latent hazard. The flat reliability curve in `data/winrate-backtest.json` needs another explanation — and note that on human corpus positions the explore=1.0 leaf is **not** flat: its ECE is 0.104 with a monotone curve running 0.166 → 0.842, against 0.196 for greedy.

4. R2 and R3 stamp their configuration now — and R3's published number has no control

`engine/run_stamp.js` is one implementation of the sidecar `rollout_r1.js` hand-rolled inline, so the next gate cannot grow a second format. It writes `<artifact>.meta.json` — the same convention that makes `data/rollout-r1-rows.meta.json` describe `data/rollout-r1-rows.jsonl` — carrying N, explore, every knob including the ones left at a default, sha256 content digests of every source the gate reaches, the commit, and **whether the tree was dirty**. A clean commit id over a dirty tree is a lie of exactly the kind this exists to stop.

Both published numbers reproduce as arithmetic, and neither reproduction is worth much.

That is the finding, and it is a different finding from R1's.

gate	published	recomputed from committed evidence	reproduces
R2	477 boards over 200 games; 5.83 ms median at n=10	the affordability table (K=3 → 0.47 s median, 1.75 s worst; K=4 → 1.49 s / 5.53 s) reproduces to the digit from <code>leafCostMs</code>	derived layer yes, base layer NOT CHECKABLE
R3	72.9% over 70 decisions (19 agreed, 20 skipped)	$100 \times (70 - 19) / 70 = 72.857142857142854$, bit-identical to the stored float	yes, and it is a tautology

R3's divergence is a pure function of two fields in the same file. There are no per-decision rows, so "it reproduces" means the artifact is internally consistent — nothing more. R2 dumps no per-leaf timing at all, and a duration cannot be recomputed by anyone in principle: it is a fact about a machine under a load, and nothing records the CPU, the node version or what else was running. **R2 is the one rung that is re-run or it is nothing.**

THE R3 RESULT IS NOT INTERPRETABLE AS PUBLISHED, and this outranks the sidecar

work. `rollout_r3.js` computes the only control that makes a divergence rate mean anything — the same search on a different seed disagreeing with **itself**, where the truth is 0.00 by construction — and it `console.log`s it and does not write it. Its own verdict branches on that number: `rate <= floor` prints NOT A RESULT. So `data/rollout-r3.json` cannot say which branch its own run took.

`docs/ROLLOUT-design.md` §5 does publish floors — 71.7 / 50.0 / 45.5 / 43.8% — but **for four earlier runs, none of them this one**. At N=20 that floor measured *higher* than the divergence. The committed artifact is a fifth run at N=600 on 70 decisions, and its floor was printed to a terminal and lost. `engine/status.js` and `docs/MILTANK.md` both quote its 72.9%, and `MILTANK.md` spends it on a decision: "so it does diverge, and the equilibrium version is worth building."

Read plainly: **the divergence is probably real** — the doc's floor fell from 71.7% to 43.8% as N rose from 20 to 200, and this run used N=600, so its floor should be lower still. But *probably* is an inference from a different run, and the Wilson interval on 51/70 is **[61.5%, 81.9%]**, which is wide enough that a 44%-class floor is the only thing separating a result from an artefact of the argmax. The next run writes the floor; until one does, the 72.9% is a headline with its control missing.

A second defect, found on the way: `data/rollout-r3.json` 's own caveat is false about the run it

describes. It reads "Switch candidates are excluded and counted". Commit `b4ec80b` deleted the `if (ca.switchTo || cb.switchTo) continue;` line — switches went **on** the menu, which is what that commit was

for — and left the string alone. It has shipped that way since 2026-08-03, and the `withSwitch / choseSwitch` counters that commit added were printed and never written, so its own headline ("4 of 12 when one is on the menu") lives in a commit message.

R2 timed a leaf the bot does not run. `rollout_r2.js` called `RL.rolloutWinProb` without `explore` or `maxTurns`, inheriting `engine/rollout_leaf.js:197`'s `explore = 0` and `engine/medicham2-browser.js:1079`'s `maxTurns = 20`. MILTANK's in-game leaf is **explore=1.0 at maxTurns=60** — a randomised payout at three times the horizon. That is R1's hole in cost form: two library defaults, written down nowhere, deciding the number. Both are now explicit, overridable and stamped, with defaults that preserve the old behaviour exactly so nothing re-dates the committed artifact by accident.

Also corrected in the generators, all of them visible in `status.js` :

- `games` was the `GAMES` environment **cap**, not a count. `status.js` printed "477 boards over 200 games", so an environment variable was being read as a measurement. It is now the distinct games actually traversed, with the cap beside it as `games_requested`.
- `leafCostMs` quantiles per N were computed over possibly-different board sets — a leaf returning null at one N and not another silently misaligns the columns — and only the n=10 count was recorded. `samples_per_n` now records all of them.
- R3's disagreement-gap median was computed twice, once to print and once to store. One variable now.
- `docs/ROLLOUT-design.md` §5's "roughly 200x the simulated turns per millisecond" is **155x** by the arithmetic of the two artifacts it cites (10×20 turns / 5.83 ms against $1 / 4.52$ ms), and 155x is itself a ceiling because it assumes no playout ends early. `rollout_r2.js` now prints the division instead of a remembered figure. **The doc still says 200x** — see filed, below.

Two retrospective sidecars were written by `node engine/run_stamp.js --reconstruct`, which infers the build from the commit that carried the artifact and marks every field `reconstructed: true`. Both score HIGH: `data/rollout-cost.json` was written 25 s before `05248f2`, `data/rollout-r3.json` 159 s before `b4ec80b`. That is evidence about a commit, not a record of a run, and it says so on every line — a stamp that hashed today's sources would describe the file rather than the run, which is `data/rollout-r1.json`'s own stated reason for recording null.

Filed, not fixed:

- `data/rollout-cost.json` **should be** `data/rollout-r2.json`. It is the only rung whose file does not carry its gate's name. Four readers: `engine/status.js:230`, `web/build-status.js:200` and `:265`, and the generated `web/status-data.js`. Three of the four are under `web/`, which MEASURE does not own. A rename that misses a reader prints NOT DERIVED and reads as "nobody ran this", which is worse than the inconsistency. Needs WEB in the same pass.
- `~~ n / n_unit` **on R1 and R4.~~ DONE 2026-08-04** — see §7 below.
- `~~ engine/rollout_r1_join.py` **writes a naked** `isoformat()` **~~ DONE 2026-08-04, and it was five files, not one** — see §8 below.
- `docs/ROLLOUT-design.md` **§5's 200x, and §R3's PASS**. Both are SEARCH's document and a SEARCH explore sweep is live. §5 should read 155x-at-most, and the R3 PASS should name which run it is quoting, because the floors in its table belong to runs that are not the committed artifact.
- `docs/MILTANK.md:70` **spends R3's 72.9% on a build decision** without its control. Same owner, same reason.
- `engine/rollout_r1.js` **should call** `engine/run_stamp.js` instead of its inline copy. SEARCH holds that file for the explore sweep. The shapes are identical today; two copies is how they stop being identical.

5. The possibly-stale artifacts, and the one class the checker could not see

`node engine/provenance.js` lists them; `node engine/provenance.js --graph` now prints the derived artifact graph itself, which is the part of that tool that could be silently wrong. Most entries are ordering artefacts inside a single run and are already annotated as such. The ones to actually chase are those older than `policy-weights.json`, those recording no game count at all, and — new — those carrying a **CORPUS DRIFT** note.

THE CANONICAL READER WAS HIDING ARTIFACTS FROM THE CHECKER. `provenance.js` derived an artifact's inputs by looking for a filename beside a read verb. A generator that loads the store the *recommended* way — `loadGames()` / `load_games()`, which resolve the path inside `engine/quality.js` and `engine/store.py` — never names `games.ladder.jsonl`, so it recorded **no dependency on the store at all** and was reported `ok` forever. Doing the right thing was the thing that made you invisible. Store derivation is now detected by the `LOADER CALL` (or an import of the reader), which is what a generator actually does.

Three more attribution faults surfaced with it, each of which had the same effect of exempting real artifacts from every corpus check:

- **A read** `open()` **looked like a write.** `pokemon-roles.json`, `role-matchups.json` and `roles-eval.json` were credited to `engine/build_roles_js.py`, which **READS** them to build a browser bundle, instead of `engine/roles.py`, which computes them from the store. `build_roles_js.py` touches no games, so all three were classed not-store-derived. A write test at line scope now requires a mode string.
- **The Python** `OUT = os.path.join(...)` **idiom hid a writer entirely.** `data/guru-matchups.json` — the source file at the centre of the `guru.js` divergence — had **no detected generator and was absent from the audit**. One level of variable indirection is now resolved.
- **Following into** `engine/quality.js` **classified everything as open-sheet.** `quality.js` names every store by construction, in its comments and in the error message that tells a caller how to pick one. `data/winrate-backtest.json`'s 6,886 **ladder** games were being judged against the 8,173-game open-sheet ceiling. It is a named exception now, with that reason.

The graph went from 76 artifacts (49 store-derived) to **84 artifacts (57 store-derived)**. One of the eight newly visible files, `data/counters.json`, was **older than the quality filter** — the **UNSAFE** condition this tool exists to catch, invisible for nine days. Regenerated (15 s); `provenance.js --strict` is green.

5g. AND IT WAS HIDING SEVEN MORE — the write detection had the same hole in the other language

84 artifacts → 91, 57 store-derived → 60, and two of the seven were UNSAFE. Found 2026-08-04 while writing the missing `docs/MODELS.md` entries: the roster guard reported `data/move-priors.json` as generated by `engine/state_encoder.py`, which only **reads** it. Three defects, each the same shape as §5's and each found by chasing the previous one.

- `const` **broke the path-indirection arm.** §5 taught this file the Python idiom `OUT = os.path.join(...)` ... `json.dump(..., open(OUT, "w"))`. The JavaScript spelling is `const OUT = process.argv[3] || path.join(...)`, and the capture took the **KEYWORD** and then failed on the `=`. Every generator using it scored zero. The cost is the §5 cost exactly: `engine/policy.js` loads the store through `quality.js`, `state_encoder.py` opens no game file, so **the behaviour clone that nine files read was classed not-store-derived and exempt from every corpus check here**. It is 2026-07-31 vintage — the 5,269-game era — and nothing could say so.
- **A READ assignment is not a writer, and accepting** `const` **proved it immediately.** `const r = JSON.parse(fs.readFileSync(...'regulations.json'...))` followed later by an unrelated `fs.writeFileSync(file, r.body)` credited `engine/fetch_smogon_stats.js` with generating the format registry — a one-letter identifier matching `\br\b` inside any later write. An assignment whose own right-hand side is a read verb now never establishes a writer.

- `named()` **was a substring test, and the comment claiming that was safe was FALSE for the most-read file in the repository.** `ladder.json` is a substring of `games.ladder.jsonl`, so every generator that opens the game store was recorded as naming `data/ladder.json` — which is how `engine/refresh-site-data.NOARCH.py` was credited with generating **MACHAMP's hill-climb artifact**, whose real writer is `engine/ladder.js` (the on-disk keys are `ladder.js` 's to the letter). The same fault hung a **phantom** `ladder.json` **input** on every store reader, and `roles.js` inside `pokemon-roles.json` did it again across eight more artifacts. Those show up as "*older than its input*" notes about dependencies that do not exist. An occurrence now only counts when the name is not the prefix of a longer one.

Corrected attributions, each verified against the generator rather than trusted: `move-priors.json` → `engine/policy.js`, `ladder.json` → `engine/ladder.js`, `dynamics.json` → `engine/dynamics.js`, `rollout-r4.json` → `engine/rollout_r4.js`, `smogon-priors.json` → `engine/smogon_priors.js`. Newly visible: `bring-bias.json`, `bring-priors.json`, `brood.json`, `core-matchups.json`, `exploitability.json`, `playstyle-matchups.json`, `smogon-priors-bo3.json`.

Two of the seven were UNSAFE, and the split between them is the point.

- `data/bring-priors.json` **was genuinely UNSAFE** — five minutes older than the quality filter, so computed under a different definition of which games count. It reads the store through `quality.js`. Regenerated (30 s), and it moved a figure a long way: `n_sides` **5,368** → **14,456**, and the format's **mega rate had been measured on 62 sides and is now measured on 12,442**, `p_side_megas` **0.9355** → **0.8785**, `p_mega_is_lead` **0.5345** → **0.5159**. `CLAUDE.md` sets a domain RATE floor there — "*a game without a mega should be rare*" — and the floor was being checked against 62 sides.
- `data/exploitability.json` **is a FALSE POSITIVE of the filter rule and a TRUE negative anyway, and it is left RED.** `engine/exploit.js` reads no game store at all — it plays self-play games from `policy-weights.json` — so the quality filter has no bearing on it, and the honest fix is a `not_store_derived` declaration, which only a re-run can write. **I did not add one, deliberately.** Stamping it would make a **genuinely unquotable** artifact look clean: it is PRIORITIES #18, WOBBUFFET's 63.2% fitted on **17 features against the 53 we ship**, and it is rendered on `web/stadium.html` and `app/stadium.html` today. Re-running it is a ~4,000-game adversarial search against a mid-flight MAG, which the engine release boundary forbids.

So `node engine/provenance.js --strict` **exits 1 on one artifact, and** `tests/run-all.js` **gates on it. That is stated, not filed.** The artifact has been invalid since 2026-07-26; the only thing that changed today is that something can see it. It needs Will's call between re-running WOBBUFFET after the release boundary and pulling the number off the two stadium pages.

5a. CORPUS DRIFT — and the answer to "two definitions of clean games"

There are not two definitions. There is one, and the other number is four days old.

`data/live.js` and `data/winrate-backtest.json` said 6,943; `data/meta-usage.json`, `data/roles-eval.json` and `data/guru-matchups.json` said ~5,269. Measured rather than argued:

figure	written	what it is
6,943	2026-08-04 03:09	<code>load_games(clean=True)</code> over the store as it stood then
6,890	2026-08-04 02:52	the same predicate, 17 minutes earlier — <code>backtest_winrate.js</code> began its run then and the collector appended exactly 53 clean games while it ran
6,886	—	6,890 minus 4 games whose <code>winner</code> matches neither player's name. A genuinely narrower question, and it is already NAMED: <code>scorable</code> / <code>dropped_no_label</code>
5,269	2026-07-31 16:42	the same predicate over a store holding 29,117 collected instead of 38,587

figure	written	what it is
5,265	2026-07-31 16:43	5,269 minus the same 4 unlabelable games

Collected grew $\times 1.325$ and clean grew $\times 1.318$ over those four days. A changed predicate does not scale with the corpus; a snapshot does. And `tests/test-quality.js` run tonight has the JS and Python readers selecting the **identical** 6,943 ids, sha `60aab8e1978e7554` on both sides. So renaming anything would have been wrong: **the defect was a date, not a word.**

Why nothing caught it: mtime cannot. The store is append-only and its mtime moves every hour, so an mtime rule would mark every store-derived artifact stale within an hour of being rebuilt — a gate that cries wolf. `provenance.js` now compares the **declared count** against the clean corpus and warns past 10%, which the measured growth rate ($\sim 7\%$ /day) makes "roughly a day and a half behind". Thirteen artifacts are flagged, including all three named above at 24.1–24.2%.

Two supporting fixes, both of which were the reason the headline artifact escaped:

- `declaredGames` now reads an explicit corpus claim first — `provenance.funnel.clean`, `provenance.usable`, `corpus.clean_games`. `data/meta-usage.json` states its population more carefully than any other file in the repository and had **no key the checker looked at**, so the file that started this question was the one it could not see a count for.
- The drift check ignores a bare `games` key, and that is deliberate: `rollout_r2.js` published `games` as the GAMES environment **cap**. Until every writer says whether `games` is a corpus or a sample, a drift figure computed on it is a guess, and the fix belongs in the generator.

Do not touch `data/quality-filter.json` **to record the new funnel**. Its mtime is `FILTER_MT`; bumping it marks every older artifact UNSAFE and turns `--strict` red across the repository.

5b. `data/guru.js` **said 0 where** `data/guru-matchups.json` **said 6 — and both were misleading** `build/build_guru_js.js` read `g.decisive`. `engine/guru.py` writes the list as `decisive_matchups`. A missing key gave `[]`, the generator then recomputed `n_decisive` from **its own empty fallback**, and shipped a provenance note asserting "ZERO statistically-decisive matchups on this population" as though it were a finding. The 144-cell matrix was byte-identical throughout, which is why nobody noticed. `venusaurmega` / `venusaur-mega`, in a new pair of files.

The true value is 6 directed = 3 distinct matchups, and the generator carries the source's count now instead of recomputing it. Three things stop it recurring, all derived: every source key must be projected or named in `DELIBERATELY_UNUSED` with a reason; the source must agree with itself (`decisive_matchups.length === min(n_decisive, 20)`); and `build_guru_js.js --check` rebuilds the bundle in memory and diffs it, run by `tests/test-guru-derived.js` on every suite run.

And the measurement underneath it: 3 of 66 pairs is what chance produces. Each cell is its own 95% test. Over 66 unordered pairs the expected number clearing that bar with no real effect is **3.3**, and **3** clear it. The smallest exact two-sided binomial p-value in the matrix is **6.1e-3** against a Benjamini-Hochberg threshold of **7.6e-4**, so **zero survive FDR at q=0.05** and zero survive Bonferroni. The bundle publishes both counts (`n_decisive`, `n_decisive_corrected`) plus the arithmetic. The old file's "ZERO decisive" string was accidentally right and arrived there by a bug — which is worse than being wrong, because it cannot be checked.

`web/index.html:1845` gates a panel headed *"These are the matchups we can actually trust"* on `GURU.decisive.length`, so it will now render three matchups that do not survive multiplicity. It should read `decisive_corrected`. WEB's file; flagged, not edited. Note the same issue already affects `isSig()` in the matrix and the "statistically significant loop" claim, independently of this fix.

`data/guru-matchups.json` is itself 24.2% behind the corpus. Regenerating it is a separate, deliberate refresh — it moves every number the GURU booth renders — and is not done here.

5c. The thirteen drifting artifacts — TRIAGED, and NONE of them is a silent refresh

`node engine/provenance.js` flags thirteen artifacts 10.6–47.2% behind the clean corpus. The question that decides what to do with each is *does regenerating it move a published figure*, and it was measured rather than guessed: every scalar in each artifact was matched against the living docs and the site pages, at headline depth, with the universal constants (0.693, 0.25, 50%) excluded because they appear for reasons that have nothing to do with the artifact.

The answer is that there are zero safe silent refreshes in the set. Nine carry a verdict string or an interval-based claim that regenerating could flip; the other four have headline figures typed into `MODELS.md`, the white paper or `SUMMARY.md`, which `engine/sanity_check.py` §5 cross-checks. By this project's own living-docs rule, regenerating any of them is a docs pass, not a refresh.

artifact	behind	what regenerating moves	act
<code>war.json</code>	47.2%	verdict <i>"WORSE THAN A COIN AT EVERY REGULARISATION STRENGTH TESTED"</i> ; <code>held_out.log_loss</code> 0.694 in <code>MODELS</code> + white paper	STOP — a null on a corpus that has since doubled is the most interesting one here
<code>policy-eval.json</code>	43.8%	verdict <i>"phase-conditioning did not help; species-only prior retained"</i>	STOP
<code>pory-eval.json</code>	33.4%	<code>log_loss.pory</code> 0.6298 in white paper + <code>SUMMARY</code> , gated by <code>sanity_check</code>	STOP — restamped instead, see §5d
<code>pory-nn.json</code>	29.4%	Blast radius OVERSTATED in this row and corrected 2026-08-04. <code>val_logloss</code> and <code>auc</code> are not keys in the file — it holds an <code>arms</code> array with per-arm <code>logloss / acc / auc</code> . And the 71.6% in <code>MODELS.md</code> and the white paper is the policy clone's top-3 accuracy , a different measurement that happens to match. No living doc cites PORY-NN , so regenerating moves zero published figures. Regenerated	done
<code>xatu-belief.json</code>	29.3%	<code>n_games</code> 4,910 and <code>top1_accuracy.belief</code> 31.2% in <code>MODELS</code> ; an improvement CI clear of zero	STOP
<code>guru-matchups.json</code>	24.2%	every number the GURU booth renders; <code>log_loss_matchup_prior</code> 0.712 in the white paper	STOP — and explicitly not in this pass, WEB is in that booth
<code>roles-eval.json</code>	24.1%	headline <i>"0.6935 vs a coin 0.6931 and rating 0.6967"</i> — a knife-edge that regeneration can flip either way; six figures in <code>MODELS</code>	STOP
<code>pokemon-roles.json</code> , <code>role-matchups.json</code>	24.1%	same generator as <code>roles-eval</code> (<code>engine/roles.py</code>); all three move together or not at all	STOP
<code>vocab-usage.json</code>	24.1%	<code>role_coverage_of_battle_usage</code> 97.2% in <code>MODELS</code>	STOP (one-line docs pass)
<code>xatu-context.json</code>	24.1%	improvement CI [0.022, 0.042] rendered on the site	STOP
<code>meta-usage.json</code>	24.1%	nothing typed — the closest thing to a clean refresh, and PRIORITIES #16 names <code>node engine/analyze.js</code>	ASK — <code>engine/mag_bot.js</code> and <code>engine/mew.js</code> read it, so it is the live bot's meta prior, and moving that is not MEASURE's call with an engine release

artifact	behind	what regenerating moves	act
		<code>data/games.ladder.jsonl</code> as its closing command	boundary pending. It is not a refit trigger: <code>engine/feature_fixture.js</code> excludes it by name and <code>board.js</code> never reads it
<code>counterplay.json</code>	10.6%	<code>result.mean_coverage_gap</code> 0.0321, CI [0.0086, 0.0563] — an interval that currently excludes zero	STOP

A false positive in the drift check itself, measured not argued. `pory-eval.json` is reported 33.4% behind, and it cannot be less than ~21% behind however often it is regenerated. Its population is not *clean ladder games*; it is *clean ladder games whose raw log is present and names a winner*, a strict subset. Running the generator over the whole current corpus reaches **5,456 games, not 6,943**, so its true drift is 15.3%. Every artifact reading `games.ladder.raw-logs.jsonl` has this. `provenance.js` 's existing escape hatches (`gate` , `games_requested` , `sampled`) do not cover it, because this is neither a gate nor a deliberate sample — the artifact needs to declare the ceiling its population can reach, in the same style. Not fixed here: `provenance.js` was built tonight and a second hand on its drift arithmetic is how two files come to disagree about one fact.

5d. PORY — the artifact restamped, and the coefficients were wrong for ten days

The verdict was not stale. The generator was answering the wrong question, correctly, every run. `data/pory-eval.json` still read *"a real, calibrated value net"* ten days after PORY was retracted, and `engine/pory.py` 's gate was `hi < coin` and `hi < material_heuristic` — which is TRUE on this sample (`hi` 0.6456 against 0.6931 and 0.6550). Restamping the file alone would have been undone by the next run. `material_heuristic` is a crude 0.75/0.25/0.5 **sign** rule; beating it is arithmetic.

The tie is now measured, not inferred. Against a logistic on `[alive_diff, hp_diff]` alone — same gradient descent, same standardisation, same temporal split — PORY scores **0.629799 to 0.629778**: paired difference **+0.000021 (POR Y worse)**, 95% CI **[−0.000013, +0.000056]** clustered by game over 925 held-out games. On the current corpus (5,456 games) it is **−0.000001**, CI **[−0.000031, +0.000030]**. The retraction is robust to the corpus growth.

The reduction is structural, so no amount of data changes it. Every state is emitted from both perspectives with the label flipped, so the gradient on any column identical across the two rows cancels exactly: intercept and `turn/10` are pinned to `0.000000000` , not shrunk to it. `my_alive` and `foe_alive` swap and come back exactly antisymmetric (sum `0.000000000`). Five features, two degrees of freedom.

`engine/pory.py` **reproduces its own artifact bit-for-bit** — replayed on the identical first-4,623 clean-game sample it returned this file's weights, `feat_std` and log-loss exactly. So the fault was never the arithmetic. The gate now reads the paired difference, the withdrawn string travels under `withdrawn_verdict` , and `reduced_form` is derived from the file's own weights.

REGENERATED 2026-08-05 on the current corpus, deliberately (the dispatch Will approved). `data/pory-eval.json` now describes **5,883 games / 97,732 board-states** (was 4,623 / a 925-game test split) and **declares** `population_ceiling: 5883` — the §5f hatch, written by the generator on its first deliberate run since the hatch existed. Everything below survives the growth: paired difference vs the two-feature logistic **+0.000001**, 95% CI **[−0.000026, +0.000029]** clustered over 1,177 held-out games; the verdict string is unchanged. The numbers a document would quote moved: log-loss **0.6298** → **0.6236** [0.607, 0.6387], sign-rule heuristic **0.655** → **0.6428**, two-feature baseline **0.629778** → **0.623623**, reduced form **0.9943 / 1.4080** → **0.9809 / 1.4093**, accuracy 0.6264 →

0.630, ECE 0.017 → 0.0138. Doc locations quoting the old figures are listed in §13b; propagation is the router's pass, not this file's.

The documented coefficients had no artifact behind them — the P1 class. 1.256 / 1.544 in docs/MODELS.md and web/stadium.html:342 is commit 44e0fb0 (2026-07-24, n_games 7,381), the last run fitted on the **unfiltered store with bot games in it.** 7f74236 put every model behind the clean filter on 2026-07-26 and the coefficients moved to 1.0259 / 1.4347, then 0.9946, 0.9962, **0.9943 / 1.4080.** The retraction has been citing bot-contaminated coefficients as its evidence ever since. MODELS.md is corrected with the history; web/stadium.html:342 **still says 1.256 — WEB's file, flagged not edited.**

5e. tests/test-site-data-fresh.js — two rules in it were wrong

It kept a second definition of stale, and it was the one provenance.js **had already rejected.** The verdict-input check compared each artifact's mtime against the newest games.*.json1 and failed past a day. The store is append-only and the collector runs hourly, so that clock cannot be beaten: five artifacts were red and **four of them are clean** by the canonical rule. It delegates to provenance.js now, the same way status.js does. The founding case survives the change — chomp-ev.json four days behind is ~28% drift, well past the 10% threshold.

What delegation loses is stated rather than dropped: drift can only see an artifact that declares a corpus, and chomp-ev.json, eval-report.json, policy-weights.json, policy-weights-joint.json and damage-validation.json declare none. They are **listed every run without failing**, so the pressure is on the generator to record a count — the shape tests/test-timestamps.js already uses.

--fix **would have refitted two models to make a freshness check go green.** The guard that stops it detected a publisher by the filename suffix -eval.json, which is not a property of anything. It caught engine/pory.py. It did **not** catch engine/nmf_roles.py (writes nmf-roles.json) or engine/xatu.py (writes xatu.json) — both fitted models quoted in MODELS.md, both on the auto-run list. The rule now is that a bundle writes only browser files and a generator that also writes a data/*.json is a publisher; checked against all ten generators this test names.

Seven of the ten stale bundles were regenerated. **Four were byte-identical apart from a date stamp** (mew.js, move-effects.js, mega-formes.js, status.js) and two entirely so (abra-meta.js, roles.js) — pure mtime, the check crying wolf. **Two had really rotted:** mag.js was serving standard errors from before the last weight change (0.02452 against policy-weights.json's 0.02363) and scoreboard.js was rendering superseded weights (1.1887 against 1.0884). That is the class this check exists for and it was real.

Also found: build_mag_data.js and build_scoreboard.js **crash without** SHOWDOWN_PATH, so --fix fails on them in any shell that has not exported it, and the test does not say so. Same shape as Po #40 — two ratchets that crashed rather than failed for the same reason.

Still red, not filed: data/pory-nn.json at **29.4% corpus drift.** The command is python engine/pory_nn.py; it is a neural-net train and it republishes val_logloss 0.612 and auc 71.6%, which MODELS.md, the white paper and SUMMARY.md all quote. That is a stop-and-ask, not a refresh.

5f. IS THE DRIFT THRESHOLD A TREADMILL? Yes, and the unit is wrong — DECIDED 2026-08-04

data/pory-nn.json was regenerated on the current corpus and tests/test-site-data-fresh.js immediately reported **CORPUS DRIFT 15.7% — declares 6,008, 7,123 clean now.** The store grew during the retrain. That is not a bug in either tool; it is what a percentage of an unbounded append-only corpus does.

The verdict: a percentage is the wrong unit, and it is not a fraction at all — it is an age. The collector runs hourly and clean games grew 5,269 → 7,123 in four days. For an artifact of age Δt , drift is $1 - n(t_0)/n(t)$, which depends only on elapsed time. A 10% threshold is therefore "about a day and a half old",

stated in a unit that hides the fact that it is a clock — which is exactly what §5e removed from `test-site-data-fresh.js` and put back through the front door. A freshly-regenerated artifact failing its own freshness check on the day it was made is the shape of a check that gets filed as *known*, and CLAUDE.md names normalisation, not invisibility, as how the docs-currency guard rotted.

The unit that answers the real question is absolute power, and `provenance.js` now prints it.

Every drift note carries a POWER line beside it:

- `ci_gain` — how many percentage points narrower the 95% interval would be. Precision goes as $1/\sqrt{n}$, so $1.96 \times 0.5 \times (1/\sqrt{n_{dec}} - 1/\sqrt{n_{now}})$.
- `max_shift` — how far the pooled point estimate could move if every missing game arrived. The pooled mean shifts by $(m/n_{now})(\bar{x}_{new} - \bar{x}_{old})$ and $se(\bar{x}_{new}) = sd/\sqrt{m}$, so a 2sd bound is $2 \times 0.5 \times \sqrt{m} / n_{now}$. Worst case, not expected case.

Measured across all thirteen drifting artifacts, the percentages span 4× and the power spans 2×:

artifact	drift	missing	CI gain	max shift (2sd)
<code>war.json</code>	48.6%	3,460	0.46 pts	0.83 pts
<code>policy-eval.json</code>	45.2%	3,220	0.41	0.80
<code>pory-eval.json</code>	35.1%	2,500	0.28	0.70
<code>xatu-belief.json</code>	31.1%	2,213	0.24	0.66
<code>guru-matchups.json</code>	26.1%	1,858	0.19	0.61
<code>roles-eval.json</code> and family	26.0%	1,854	0.19	0.60
<code>pory-nn.json</code>	15.7%	1,115	0.10	0.47
<code>counterplay.json</code>	12.8%	915	0.08	0.42

No artifact in this repository has enough missing data to move a proportion by one percentage point. `war.json` is missing *half its corpus* and can move 0.83 points. The smallest split-half noise floor this division has published is **0.43 points** (R1's cuts run 0.43–2.01; R4's three run 0.2 / 1.3 / 3.9), so `counterplay.json` is already **below the noise floor** and the rest sit inside a factor of two of it. "24% behind" and "the games it lacks cannot move it past its own noise floor" are different statements and only the second one is actionable.

And it self-extinguishes, which is the property the percentage lacks. `max_shift` = $\sqrt{f}/\sqrt{n}$, so the same 15.7% drift that moves 0.47 points at n=7,123 moves 0.33 at n=14,000 and 0.24 at n=28,000. The treadmill stops on its own as the corpus grows instead of being switched off by hand.

What was NOT changed, deliberately: the 10% trigger. Two reasons, and the second is the honest limit of this work.

- Lowering the bar changes thirteen artifacts' status and that is not a call to make inside a measurement pass.
- `max_shift` **still cannot see the thing that decided every row of §5c's hand triage** — the DISTANCE from an artifact's headline estimate to its decision boundary. `roles-eval.json` publishes 0.6935 against a coin's 0.6931; that 0.0004 margin is flippable by any new data at all, while `war.json`'s null is not flippable by 0.83 points. That margin is not computable from `n`, and the artifact is the only thing that knows it. `max_shift` is also stated for a **proportion** at sd = 0.5; a log-loss lives on another scale and the number is not directly comparable there. The next rung is a declared `decision_margin`, in the same convention as below.

A grace period measured in regenerations is rejected. It is a clock with extra steps, it cannot tell `chomp-ev.json` from `pory-nn.json`, and a fixed window is a licence for a genuinely flippable artifact to sit quiet inside it.

The `pory-eval.json` false positive is fixed on the reader side and still needs its generator.

`provenance.js` now honours a declared `population_ceiling` (`j.population_ceiling`, `provenance.population_ceiling` or `corpus.population_ceiling`) and measures drift against it — the same declaration convention as `not_store_derived`, `raw_store_ok`, `gate` and `games_requested`. `pory-eval.json` is a strict subset (clean ladder games whose raw log exists AND names a winner: 5,456, not 7,123), so it can never get below ~21% against the wrong denominator. **This is a separate defect from the unit question and the answer above does not fix it by itself:** the hatch exists, and `engine/pory.py` must write the key on its next deliberate run. **DONE 2026-08-05** — the deliberate run happened (§5d addendum) and the generator now writes `population_ceiling` with a note naming its predicate. Every artifact reading `games.ladder.raw-logs.json1` has the same shape.

WHAT THE NEW RULE WOULD AND WOULD NOT HAVE CAUGHT, plainly.

- `data/counters.json`, **older than the quality filter, UNSAFE for nine days — CAUGHT, and it was never a drift case.** That is the `FILTER_MT` check: the PREDICATE changed, so the artifact answers a different question, and no amount of power makes it valid. It is untouched, it is still `bad` rather than `warn`, and it still fails `--strict`. Confusing the two checks is how a volume rule gets credit for a correctness rule's catch.
- `data/chomp-ev.json` **four days behind, publishing "does not beat a coin" about a model with a directional edge — CAUGHT, and better than before.** That verdict sits *at* its boundary, so its decision margin is ≈ 0 and any `max_shift` exceeds it. The percentage caught it at 28% > 10%; the power rule catches it for the right reason.
- **An artifact recording a corpus it did not use — NOT CAUGHT, by either rule, and this file already says so on every run.** Only re-running the generator can.
- `data/slowking-playstyle.js`, **a GURU run written under the playstyle name — NOT CAUGHT by anything, and this is why the crude mtime rule in `test-site-data-fresh.js` was left alone.** See §9.

6. The noise floor is not a standing artifact

Split one arm in half and measure the spread. An effect smaller than that is not an effect. This gets re-derived by hand every time somebody needs it, which means it usually is not derived at all.

Two consumers now emit their own and neither is general: `rollout-r4.json` carries three split-half cuts of the H2H, and every block of `winrate-backtest.json` carries a `noise_floor` on Brier and on accuracy. That is the right shape — the floor belongs to the measurement, not to a global constant — but there is still no A/A run for the H2H, and a floor computed inside the arm being judged cannot see between-run variance.

7. All four rungs carry `n_measured` / `n_unit` — CLOSED 2026-08-04

`engine/rollout_r1_artifact.js` and `engine/rollout_r4.js` now write the pair R2 and R3 already carried, and both artifacts were regenerated from committed evidence (no rollouts): `data/rollout-r1.json` **9,201 scored positions**, `data/rollout-r4.json` **535 decisive pairs**. Choosing which of R4's four numbers goes in the common slot is the whole point of having one — the SPRT is computed on decisive pairs and nothing else, and the handoff quoting "5,248 games" (the line count of a store that writes two lines per game) is what the slot is for.

Still **not** called `n`: `data/rollout-r3.json` has published `n` as the rollout BUDGET since 2026-08-03, and one key meaning a sample size in one rung and a budget in the next is worse than no common key.

`tests/test-rollout-gates.js` derives the rung list from the filenames `engine/rollout_r*.js` write, asserts every generator emits both keys, and then permits exactly one artifact state beyond "carries them": *its generator does, awaiting a re-run*. `data/rollout-cost.json` is in that state and cannot leave it here — it is a set of TIMINGS, and R2 is re-run or it is nothing. What the test forbids is the state that actually goes wrong: missing in the artifact **and** in the generator, which is nobody having done it.

8. Naive timestamps — CLOSED 2026-08-04, and it was FIVE writers, not one

`engine/rollout_r1_join.py` was the reported case. The real answer to "is one occurrence a typo or a pattern" is that `datetime.now().isoformat(timespec="seconds")` appeared in **five** generators — `rollout_r1_join.py`, `lookahead_bound.py`, `lookahead_clock_control.py`, `nmf_rank.py`, `polygon2.py` — which makes it the house style rather than a slip. Eight committed artifacts carry one, and all eight come from exactly those five.

Correct the diagnosis, not just the bug. JavaScript does not misparse it. ECMA-262 gives the two ISO forms opposite defaults — date-TIME with no offset is read as LOCAL, date-ONLY is read as UTC:

```
new Date('2026-08-03T04:14:10') -> 2026-08-03T08:14:10.000Z (local, this box is UTC-4)
new Date('2026-08-03')          -> 2026-08-03T00:00:00.000Z (UTC)
```

So the four-hour figure is the RENDERED string, not the parse, and on this machine the value round-trips. The defect is that the stamp means something different to every reader, that the two forms this project already uses side by side follow opposite rules, and that it is wrong by the reader's UTC offset the moment it is compared against a `Z` stamp — which is what every JavaScript writer here emits and what `status.js` and `provenance.js` exist to do.

`engine/isotime.py` is the single home (`utc_now()`, `utc_today()`); all five call it. `data/rollout-r1-withdrawn-join.json` is deliberately NOT regenerated — it is a withdrawn result kept so the withdrawal can be checked. `tests/test-timestamps.js` gates the WRITERS, asserts the two ISO forms really do disagree on the running machine rather than quoting a comment about it, and lists the artifacts still carrying a naive stamp without failing on them, because an artifact is fixed by re-running its generator and that pressure is how "KNOWN FAILURE" gets typed.

9. The two "stale bundles" — one was a no-op and the other was never the file it claims to be

Both were regenerated with the verify-before-trusting step first. That step is the entire finding.

`data/engine-data.js` — **BYTE-IDENTICAL. Nothing was landed.** `SHOWDOWN_PATH=...` `node build/rebuild_sets_from_sheets.js` reports 318 species, 195 rebuilt from real sheets, 123 left alone under 10 sheets, **materially changed 0, illegal abilities fixed 0**. Run with `--write` and diffed against a preserved copy: **identical to the byte**. The generator reproduces its own artifact and the 0.9-day staleness was `mtime` and nothing else.

The original `mtime` was then RESTORED, and that is the point of the entry. Writing the identical file moved `engine-data.js` forward and immediately turned `counterplay.json`, `scoreboard.js` and `winrate-backtest.json` — this division's own leaf-calibration artifact — into "older than its input `engine-data.js`". Three false staleness flags manufactured by a regeneration that changed nothing. A restamp with negative information content is still a restamp; `status.js` 's refit edge is hash-based (`feature_fixture --check`) and was never at risk, but `provenance.js` 's input-ordering rule is `mtime`-based and was.

REPAIRED AND LANDED 2026-08-04. The section below is kept as the diagnosis; this is the result. Run with **both** variables set — `TAG=playstyle MATRIX_FILE=data/playstyle-matchups.json` — every figure predicted below reproduces to the digit: `n_games` 5,265 → **2,860**, archetypes 12 → **8**, mixture → **Rain 0.8079 / Setup 0.1657 / FakeOutBalance 0.0255**, greedy-Nash 0.0409 [−0.0001,

0.1735] → **0.026 [-0.0001, 0.1498]**, uniform 0.0761 → **0.0338**, triples 1,320 → **336**, cycle → **TailwindOffense** → **Sand** → **TrickRoom** (legs on 40, 5 and 140 games, still supported: false), and **the verdict flips** to "no material exploitability gap ... close to transitive at this granularity." It reproduces itself on a second run, byte-identical, and is no longer byte-identical to data/slowking.js. **The GURU arm was re-run first and reproduces its own artifact bit-for-bit** — every shared key unchanged, which is what licensed trusting the playstyle run from the same code. **THE FIX IS THE DEFAULT, NOT THE FILE.** engine/slowking_preview.py now REFUSES to write a TAG-named artifact from the default matrix and prints the two-variable command. The rule is narrow on purpose: a TAG names a NON-default run, so TAG-set-with-default-matrix is the one combination that cannot mean anything; the ordinary GURU run and the correct playstyle run are both untouched. A relative MATRIX_FILE now resolves against the repo rather than the shell's cwd — the documented command only worked from the repository root, and a path that works from one directory and not another is how the wrong matrix gets reached for. **A second half of the same bug, found on the way:** source_matrix was the hardcoded literal "data/guru-matchups.json". So even a CORRECT playstyle run would have stamped the GURU matrix as its source — the one field that could have exposed the clobber was pinned to agree with it. It is derived now, and tag is recorded beside it. **What moves on the page** (app/index.html:907-923 / web/index.html , WEB's files, not edited): games 5,265 → 2,860; the cycle legend from three species pairs to TailwindOffense → Sand → TrickRoom; leg edge 10% → 5%; the mixture chips from Gengar-Incineroar 66% / Charizard-Garchomp 22% / Pelipper-Archaludon 12% to **Rain 81% / Setup 17% / FakeOutBalance 3%**; greedy exploitability 4% → 3%. **And two TYPED literals in that paragraph are now wrong on both pages** — "these matchups rest on 49, 37 and 15 games" (really 40, 5 and 140) and "the strongest of 1,320 candidate triples" (really 336). **Worse than the numbers: the room's whole thesis is now contradicted by its own artifact.** The panel is headed "The meta looks like rock-paper-scissors" and argues "picking one single playstyle is exploitable while a mixture isn't — the reason to mix", while the artifact it renders now says mixing buys little here. That is a WEB pass, and it is a rewrite rather than a number swap. **Three consumers checked.** engine/sanity_check.py passes, and its \$5 check "site mixture top == report top" now reads **Rain == Rain**; before the repair it compared two copies of the same wrong file and passed for that reason. tests/test-docs-current.js \$1b likewise now reads the real playstyle artifact (0.026, CI[-0.0001, 0.1498]) where it had been reading GURU's numbers under the playstyle name — a guard built to track this artifact was tracking the other one. And engine/build-status.js:18 reads slowking-playstyle-eval.json into a variable ex that **nothing in the file ever uses**; it is a consumer in name only.

data/slowking-playstyle.js — **STOP. It is not stale; it is the wrong file, and has been since 2026-08-03 15:15.**

engine/slowking_preview.py takes its OUTPUT NAME from TAG and its MATRIX from MATRIX_FILE, which defaults to data/guru-matchups.json. Run with TAG=playstyle and MATRIX_FILE unset it writes a GURU result under the playstyle name. Measured:

- data/slowking-playstyle.js has a payload **byte-identical** to data/slowking.js;
- data/slowking-playstyle-eval.json is a **byte-identical file** to data/slowking-eval.json;
- both read 5,265 games / 12 species-pair archetypes / 1,320 candidate triples — GURU's shape.
data/playstyle-matchups.json holds **2,860 games over 8 playstyles.**

Regenerating it correctly moves published figures, so it was **restored and not landed**: n_games 5,265 → **2,860**, archetypes 12 → **8**, mixture Gengar-Incineroar 0.66 / Charizard-Garchomp 0.22 / Pelipper-Archaludon 0.12 → **Rain 0.81 / Setup 0.17 / FakeOutBalance 0.03**, greedy-Nash 0.0409 [-0.0001, 0.1735] → **0.026 [-0.0001, 0.1498]**, uniform 0.0761 → **0.0338**, cycle Charizard-Garchomp→Kingambit-Garchomp→Incineroar-Whimsicott → **TailwindOffense**→**Sand**→**TrickRoom**, triples searched 1,320 →

336, and the verdict string flips from *"substantially less exploitable... the meta is non-transitive here (rock-paper-scissors)"* to *"no material exploitability gap between Nash and greedy — this meta is close to transitive at this granularity."*

The corrected numbers are the ones `docs/MODELS.md` **already publishes** in the MACHAMP entry — 336 triples, a leg on 5 games, 0.026, `[-0.0001, 0.1498]` — to the digit. So the docs are right and the artifact is wrong, which is the rare direction, and the 2026-08-02 withdrawal of the SLOWKING cycle still rests on measured evidence. `engine/build-status.js:18` and `engine/sanity_check.py:32` both read the clobbered file, and `app/index.html:907-923` renders its mixture and cycle legs. The repair is one command with **both** variables set; it is a WEB pass, not a refresh. The generator should refuse to write a `TAG`-named artifact from the default matrix.

Two things this costs the checkers, and both are recorded rather than fixed here.

- `provenance.js` reports `slowking-playstyle.js` as `ok`, correctly by its own rules — it is co-generated with `slowking.js`, so the ordering carries no information. Provenance sees ordering and declared counts; it cannot see that a file's CONTENT came from the wrong input. This is why the crude mtime bundle rule in `tests/test-site-data-fresh.js` was **left alone** despite §5f: delegating it to drift today would have marked this file clean.
- `tests/test-site-data-fresh.js` printed the repair as `node engine/slowking_preview.py` — the wrong interpreter, in the STALE table only; the `--list` path already derived it from the extension. Fixed. The command it names is still incomplete, which the test's own comments already admit, and running it with `TAG` set and `MATRIX_FILE` unset is a plausible route to the clobber that is on disk.

10. `train_value.py` was discarding a fifth of the corpus — PRIORITIES #13, FIXED 2026-08-04

`idn()` normalises punctuation and nothing else, so the event stream's `charizardmegay` never matched the bring list's `charizard`, `side_of` returned None, and the event was thrown away without a word. **Measured on 4,000 clean games before the fix: 21.7% of faints, 22.7% of damaging events, 20.8% of all damage, at least one discard in 96.5% of games, and 97.6% of discarded targets are megas.** The visible symptom is the one to remember: **88.9% of clean games ENDED with both sides still holding bodies.** The value net was learning from trajectories in which almost nobody ever loses their team.

The fix is a verb on the one resolver, not a fourth copy of it. `engine/mc_key.js` gained `mcKey.base` (which body is this) and `mcKey.bases` (the whole map, for a caller that cannot call into JavaScript per name), reading the `base` field the generator already wrote from the dex into `MC.mons`. Three properties were deliberate:

- **It is not a string strip.** `re.sub(r'mega[xy]?$', '', s)` is the obvious three lines and it is wrong — `mc_key.js` already records that the identical JavaScript strip answered Victreebel for Victreebel-Mega. The table carries the answer; this reads it.
- **It returns a flat BODY id, not a table key.** `MC.mons` holds `floette-mega` with `base: "floette"` and holds no `floette` row, so a version resolving the base back through the table returned null for exactly the 1,613 events this exists to rescue, one layer down. Being in our damage table is a fact about our table; the body is a fact about the game.
- **It touches no dex, so it cannot need `SHOWDOWN_PATH`** — the crash-instead-of-fail mode PRIORITIES #40 records for two other ratchets. `train_value.py` shells it once per run and **fails loudly** if it cannot, rather than reverting to the behaviour above.

After: 22.7% → 1.7% of damaging events dropped, 21.7% → 1.5% of faints, and games ending with both sides intact 88.9% → 26.3%.

What it moved, and the honest size of it. Paired on identical held-out states — 1,445 games, 10,120 states, both arms fitted on the same split:

	before	after	paired difference
log-loss	0.6634	0.6520	-0.0114, 95% CI [-0.0183, -0.0041] (bootstrap clustered by game)
accuracy	59.72%	61.47%	+1.75 pts, 95% CI [0.50, 2.94]

The mechanism is legible in the weights: `hpDiff` moved **0.169** → **0.377**, because a fifth of all damage had never been applied and the feature was attenuated toward zero. The shipped artifact (`ladder + self-play`, as `main()` defaults) moved `test_logloss` **0.6638** → **0.6536**.

Both intervals clear zero and the effect is still inside the noise floor for an unpaired comparison. Twenty split-half cuts of the fixed arm alone spread by a **median 1.87 accuracy points** (range 0.30–5.37) against a 1.75-point effect. The pairing is what buys the resolution; two runs on different samples could not tell these value nets apart. And the ceiling is unchanged — 61.4% sits below the **66.92%** in-sample ceiling for this feature class and below the live leaf's **67.97%**. **This is a correctness fix, not a capability change, and it was worth making on the first ground alone.**

Residual, measured rather than assumed: 1.7% still drops, and 1,613 of the 1,625 are one species. `MC.mons` carries `floette-mega` → `floette`, the store's bring lists hold `floetteeternal`, and no `floette` row exists — the chain does not close. The rest are in-battle formes the mega table does not cover (`mimikyubusted` 274, `morpekohangry` 48, `castformsnowy / castformrainy` 11). **Filed to ENGINE, not patched here:** closing them means reaching for the Showdown dex, which would make `train_value.py` produce different numbers depending on whether `SHOWDOWN_PATH` is set. That is *fitting environment and playing environment must match* in a new place, and it is not worth 1.5% of events.

11. THE THIRD COPY OF THE WEATHER MAP IS IN `board.js`, IT IS WRONG, AND THE FIXTURE CANNOT SEE IT

LANDED 2026-08-04, with the two fixture boards it needed, and refitted. The diagnosis below is kept because it is the reason the fixture grew; the result is here. `engine/board.js` no longer keeps a weather map. Both reads — `dmgFractions` and the `punishExposure` call in `featuresFor` — go through a `weatherKind(board, D)` helper that calls the damage engine's exported `weatherId`, the same consolidation ENGINE made in `tag_dex.js`. A damage engine that cannot answer is counted in `dmgFailures.weatherUntranslated` rather than defaulted to clear skies; measured over 234,873 candidate vectors it is **0**, on both the node and the browser export path (`tests/test-board-browser.js` : 58 of 58 features agree to 6 dp). **THE PRE-LANDING MEASUREMENT REPRODUCES ON THE CURRENT ENGINE TO THE ROW.** Re-run before relying on it, because the engine had moved underneath it: `fit_policy.decisionsFor` over the first 1,200 open-sheet corpus games, **32,054 decisions / 234,873 candidate vectors**, one process holding both builds of `board.js` : | | measured 2026-08-04 (pre) | re-measured after the ENGINE band | |---|---|---| | candidate vectors that move | 1,768 (0.75%) | **1,768 (0.75%)** | | decisions that move | 892 of 32,054 (2.78%) | **892 (2.78%)** | | columns that move | 14 of 58 | **14 of 58** | | games containing a moved vector | not measured | **238 of 1,200 (19.83%)** | The 19.83% is the new number and it is the one that matches the census: 18.5% of games carry sand or snow at some turn. **THE FIXTURE NOW SEES IT — 0 columns before, 10 after.** `sand-is-up` and `snow-is-up` join `SCENARIOS`. `Tyranitar` takes special hits under sand (the Rock special-defence 1.5x) and `Ninetales-Alola` and `Weavile` take physical hits under snow (the Ice defence 1.5x); `Hippowdon` and `Ninetales-Alola` both carry Weather Ball, whose TYPE resolves off the same field, so a translation that mapped one weather and not the other cannot pass both. A Rock body and an Ice body sit on each bench, because the switch family prices a body that is not on the field and would otherwise be untouched — that one choice took the detection from 6 columns to 10. Scored against the pre-fix map:

koTarget , dmgFrac , killIsRoll , killsThreat , koFirst , switchSurvives1 , switchKOFast , switchDiesFirst , benchRisk , and the joint partnerCoversMe . **State the limit rather than the win: the fixture catches 10 of the 14 columns the corpus moves.** protectThreatened , diesBeforeMoving , screenValue and switchKOSlow move on corpus boards and not on these ten. A fixture is evidence for the restamp rule, never proof of it, and this is the measured size of the gap rather than a caveat in prose. Coverage did not regress: 40 slots, 324 candidates, 1,309 pairs, and **o features that never fire**. Two properties of the landing worth keeping. weatherId still does not know desolateland / primordialsea ; on 339,483 corpus turn-boards that costs nothing and adding them is ENGINE's call on SD2WEATHER , not a fourth table here. And adding scenarios re-stamps every hash, so it went in the same pass as the refit — a fixture change and a restamp cannot be separated.

This is a refit trigger. It is measured, it is NOT landed, and the gate for landing it is the Po/P1 band, which is not met.

engine/board.js:1190 carries WEATHER_KIND , a third private copy of the Showdown-weather → engine-weather translation that medicham2-browser.js owns as SD2WEATHER / weatherId . It is read at two sites — dmgFractions (:1247 , every damage-derived MAG feature) and the punishExposure call in featuresFor (:2937 , clickCost). What it holds:

```
{ sunnyday: 'sun', desolateland: 'sun', raindance: 'rain', primordialsea: 'rain' }
```

It maps the two weathers this format cannot produce and misses the two it does. desolateland and primordialsea are primal weather: **0 occurrences in 339,483 corpus turn-boards.** sandstorm and snowscape are not in the table at all, so WEATHER_KIND[board.weather] is undefined , || '' makes it clear skies, and **every damage feature under sand or snow has been computed in no weather**. The engine reads field.weather === 'sand' for the Rock special-defence 1.5×, === 'snow' for the Ice defence 1.5×, and Weather Ball's type comes off the same field.

Exposure, re-measured rather than inherited — a census of board.weather at every turn-board across games.ladder + games.bo3 + games.ots (52,441 games):

weather at a turn-board	share	WEATHER_KIND gives
clear	64.15%	'' correct
sunnyday	14.90%	'sun' correct
raindance	10.23%	'rain' correct
snowscape	5.43%	'' — WRONG
sandstorm	5.29%	'' — WRONG

10.72% of turn-boards, and 18.5% of games contain at least one.

THE FIXTURE PASSES ANYWAY, AND THAT IS THE FINDING. The patch was applied, measured and reverted. engine/feature_fixture.js --check returns feature semantics OK on both policy-weights.json and policy-weights-joint.json **before and after** — because the fixture's only two weather scenarios are RainDance and SunnyDay , the two WEATHER_KIND already gets right. All 58 columns are hash-identical while the feature function has moved.

What actually moves, measured on the fit's own rows — fit_policy.decisionsFor over the first 1,200 open-sheet corpus games, **32,054 decisions / 234,873 candidate vectors**:

candidate vectors that move	1,768 (0.75%)
decisions that move	892 of 32,054 (2.78%)
columns that move	14 of 58

`dmgFrac (1,182)`, `koTarget (238, max  $|\Delta|$  0.93)`, `killIsRoll`, `killsThreat`, `diesBeforeMoving`, `benchRisk`, `koFirst`, `protectThreatened`, `switchKOFast`, `switchKOSlow`, `screenValue`, `switchDiesFirst`, `switchSurvives1`, `switchSurvives2` — several flipping a full 0→1.

So `feature_fixture --check` is necessary and not sufficient, and `status.js`'s `refit edge: CLEAN` inherits that limit. The hash covers the boards the fixture builds; the feature FUNCTION lives over every board the corpus contains. This division's own rule — *a restamp is only valid if the feature FUNCTION is unchanged* — is the binding one, and a green fixture is evidence for it, not proof of it. **The fixture needs a sand board and a snow board.** Adding them is itself a fixture change that re-stamps every hash, so it belongs in the same pass as the refit, not before it.

The patch is small and is recorded here rather than left on disk: replace both reads with a `weatherKind(board, D)` helper that calls the damage engine's exported `weatherId`. Do not write a fourth map. Note that `weatherId` does not know `desolateland / primordialsea`; on the measured corpus that costs nothing, and adding them is ENGINE's call on `SD2WEATHER`, not a second table here.

11a. `position_features.js` — the same defect at the second boundary. LANDED, no refit owed.

`engine/position_features.js:292` built its field object as `B.norm(board.weather || '')` — the board's *move name* — and handed it to `M.dmgRange` and `M.effSpeed`, which compare against the engine's words. Truthy and meaningless, exactly as at the leaf boundary. Now `M.weatherId(...)`.

No refit is owed and that was checked, not assumed: nothing in the repository fits, reads or renders these columns. The only callers of `positionFeatures` are four tests, and no artifact contains any of its feature names.

Exposure re-measured on the boards this module is actually asked about — the `joint_rows.build` walk, 400 open-sheet games, **3,202 mid-game boards / 6,404 scored positions. 49.94% carry a weather**, higher than the 35.85% turn-board figure because weather accumulates as a game runs.

- **1,962 of 6,404 positions moved (30.64%):** sun 73.6%, rain 66.7%, sand 49.5%, snow 29.5%.
- 7 of 16 columns: `raceEdge` (29.67%, max $|\Delta|$ 0.42), `killFirstEdge`, `iKillNext`, `theyKillNext`, `benchAnswersDiff`, `speedEdge`, `pinnedDiff`.
- **0 of 3,206 clear-weather positions moved** — the control that says this is the weather.

`:296`'s `terrain` now goes through `M.terrainId` as well. That one is a **confirmed no-op**: 16 of the 3,202 boards carry a terrain, 11 of them under clear weather, and all 22 positions scored on those are bit-identical. Every downstream reader already calls `terrainId` itself. It is translated here so the field object leaves in one vocabulary rather than two, which is the condition that let the weather half go unnoticed. The four probed keys are a list of BOARD KEYS, not a translation table — the same justification `rollout_leaf.terrainOnBoard` records.

ENGINE's "*o of 400 boards*" for this file was terrain-only and is reproduced (0.50% here). It was being read as though it covered the weather too, and the weather number is 49.94%.

11b. `git checkout -- <file>` INVALIDATES EVERY CONTENT STAMP ON THIS MACHINE

Found by doing it. `core.autocrlf=true`, the committed blobs are LF, and the worktree files are LF — so a checkout REWRITES them as CRLF. The bytes change, nothing in git notices (`git diff` is empty), and `sha256(worktree)` moves. `winrate-backtest.json`'s `measured_against` and `run_stamp.js`'s `source_digests` both hash worktree bytes, so `engine/board.js` immediately began printing `PRE-CHANGE` — measured against a different build of: ... `engine/board.js` after a revert that changed no code. Converted back to LF; the digest returns to `bcf2dab9dc6f`, which is the stamp exactly, and `board.js` drops out of the `PRE-CHANGE` list.

This is the mirror of the warning already in *Reading a stamp — never compare source digests to git.blobs*, they differ by line-ending translation. The new half is that an ordinary git operation can move one of them. **After any git checkout -- or git stash pop on this box, check node engine/status.js before believing a PRE-CHANGE line.**

12. tests/test-no-silent-failure.js — the 32 MEASURE entries, and what two of them were hiding

Worked through without --update, which would have laundered the SEARCH, OPS and WEB entries in the same command. **NEW since the baseline: 52 → 20.** Every MEASURE-owned entry is cleared; the 20 that remain are 13 SEARCH (miltank.js, rollout_leaf.js, rollout_r1.js), 3 OPS (mag_bot.js) and 4 in tests/test-{site-data-fresh, stadium-roster, web-parses}.js.

Two were hiding something real.

engine/run_stamp.js:92 recorded "the tree was clean" whenever git refused to answer.

```
const porcelain = git(['status', '--porcelain', '--'].concat(watched)) || '';
```

git() returns null when the command throws — an index lock, an interrupted rebase (CLAUDE.md documents this repository reaching one 43 commits into a 45-commit replay), git not on PATH. An EMPTY porcelain is git's way of saying CLEAN. || '' collapsed the two, so a stamp written while git was unavailable published dirty: false beside a commit id that described nothing on disk — and this module's own header says "a clean commit id over a dirty tree is a lie of exactly the kind this module exists to stop", while docs/MEASURE.md's *Reading a stamp* tells every reader to trust the commit when dirty is false. rev-parse HEAD was already guarded; status --porcelain was not. Now a third state: dirty: null with git_errors, and status.js renders it as **DIRTINESS UNKNOWN** rather than as the clean case.

engine/backtest_winrate.js:71 + engine/status.js:176 composed into a false clean bill. stampOf returns {mtime: null, error} with no sha256_12 when a source cannot be read. status.js then did if (!st || !st.sha256_12) continue; and, finding nothing in moved, printed "CURRENT — every engine source the leaf reads still hashes to what it was measured against". With every stamp failed, that sentence was printed over **zero comparisons**. Two silent catches, neither wrong on its own, producing a clean provenance line on this division's headline number. The count is now stated (all N engine sources), unstamped sources are named, NOT DERIVED is printed when N is zero, and a source that has been DELETED is reported as gone rather than as "a different build of".

The rest were latent rather than active, and the honest answer is that they were hiding nothing today — which is a measurement, not an absence of one.

- engine/rollout_r4.js:279 — the split-half scan that produces the NOISE FLOOR discarded a torn line in silence, while countLine, reading the same file for the header counts, keeps torn and publishes it. A row lost here shrinks an arm and moves the spread, and the spread is the entire output. Counted; the split now refuses to report if anything was lost. **Measured on games.r4-decided.jsonl: 0 torn, 0 bad-seed, across all three cuts.** Before the counter existed, 0 and 500 looked identical from there. A second hole found while adding it: a non-numeric seed put every such record on side B, because NaN % 2 !== 0 — now rejected rather than piled up. Incidental: the seed hash parity cut splits 1,382 / 1,242, an 11% imbalance, and it is the cut that produced the largest of the three spreads (3.9 pts) quoted as this run's noise-floor range.
- tests/test-timestamps.js:49/54 — a directory that would not list and a file that would not read were both skipped with continue, so "no Python generator writes a naive datetime" was also the answer when **zero generators had been looked at**. That is CLAUDE.md's *a capability that cannot prove it ran* inside a

guard written for a different failure. Now asserts a floor on files scanned (39 in `engine/`, 1 in `build/`; `tools/` has no `.py` and `scripts/` does not exist) and fails on anything skipped unread.

- `tests/test-timestamps.js:92` — an artifact that would not parse was silently excluded from the published "*N artifacts still carry a naive stamp*" list, so the files most likely to be broken were the ones the survey could not see. Now counted and named: **0 of 108** `data/*.json` **fail to parse**, so the list of 8 is complete — a statement that could not previously be made at all.
- `tests/test-web-status.js:181` — `catch { return false }` on the freshness filter means "not newer than the board", the same answer a perfectly fresh artifact gets. A source that had been **deleted or renamed** read as up to date, in the test whose job is that every rendered figure traces to an artifact. Missing is now its own failure. None are missing today.
- `tests/test-rollout-gates.js:81` and `engine/rollout_r1_artifact.js:228` — both collapsed "no such file" into "will not parse". The first then granted a CORRUPT gate artifact the one tolerated state ("*awaiting a re-run*"); the second made a broken sidecar indistinguishable from a run nobody ever stamped, which is the exact distinction §4 and §7 exist to preserve.
- `engine/status.js` (9), `engine/rollout_explore_sweep.js` (3), `engine/rollout_r1_artifact.js` (3 more), `engine/run_stamp.js:60`, `tests/test-web-status.js:58/112`, `tests/test-guru-derived.js:56` — each conflated *absent* with *unreadable*. `status.js` now carries a `DIAGNOSTICS` block, printed on screen and deliberately **outside** the section bodies so `--write` never stamps a transient into a ledger.

Two defects in the ratchet itself, filed not fixed:

- `--update` **is all-or-nothing, so the tool's own guidance cannot be followed.** It says "*if a silent fallback is genuinely right here, say why in the code and re-baseline with --update so the exception is deliberate and visible*" — but `--update` re-baselines every silent catch in the repo, including other divisions'. There is no way to bless ONE. It needs a per-entry allow with a reason string, in the shape `build_guru_js.js` 's `DELIBERATELY_UNUSED` already uses.
- `isSilent` **cannot see a recorder it does not recognise by name.** `error: e.message` inside a returned object is a colon, not an `=`, so an artifact that carries its own reason still reads as silent; and a named helper that pushes onto a list looks like nothing from inside the catch body. Four surviving entries are this. Widening the regex would launder real ones, so the code was moved to the documented convention instead — `status.js` 's recorder is named `logUnreadable`, and two locals are named `errWhy` / `errBundle`.

`--all` was added to the ratchet: the 25-line cap is right for a gate, but "*... and 27 more*" is how the tail of a list stops being anybody's job.

13. THE REFIT RAN — and it moved nothing measurable. That is the result, not a preamble to one.

`node --max-old-space-size=4096 engine/fit_policy.js` then `engine/fit_joint.js`, on the weather landing in §11. **8,759 clean open-sheet games, 229,339 usable decisions** (up from 8,414 / 220,613), 183,679 train / 45,660 held out, lambda selected on held-out likelihood at 0. Both weight files carry a fresh `featureHashes` over the 10-scenario fixture, and `feature_fixture --check` exits 0 on both.

The before/after in the artifacts is not the comparison to read, because the two fits have different corpora and different held-out sets — 44,033 decisions against 45,660. Quoting `heldOut.boardAware` 32.269% against 32.271% would be comparing two samples, which is the confound this division keeps finding in other people's work. The comparison that means something scores the SAME held-out decisions three ways, with the split reproduced exactly (`hash(game) % 5 === 0`):

arm	what it is	logL/decision	top-1
A	old weights + old features	-1.732548	32.204%

arm	what it is	logL/decision	top-1
B	old weights + NEW features	-1.732200	32.252%
C	NEW weights + NEW features — what ships	-1.732276	32.178%

1,772 held-out games, 46,162 decisions, paired per decision, bootstrapped over 10,000 resamples of GAMES (decisions inside one game share a team and a board):

paired difference	logL/decision	top-1 points
B – A the weather fix alone, weights frozen	+0.000348 [0.000075, 0.000623]	+0.048 [0.009, 0.093]
C – B the refit, given the fixed features	-0.000076 [-0.000172, +0.000021]	-0.074 [-0.155, +0.004]
C – A everything, against what shipped	+0.000273 [-0.000010, +0.000556]	-0.026 [-0.117, +0.064]

Read plainly, three statements:

- 1. The correctness fix is detectable and it is a quarter of the noise floor.** B – A clears zero on both metrics, and twenty split-half cuts of arm C alone spread by a **median 0.192 top-1 points** (range 0.005–0.770) against an effect of 0.048. The pairing is the whole reason it resolves at all; two runs on different samples could not tell these builds apart. Same shape as §10's value net, one order of magnitude smaller.
- 2. Refitting the weights on the corrected features bought nothing.** C – B contains zero on both metrics and its point estimate is NEGATIVE. Only **1 of 58 weights moved more than 2 SE** (`dmgFrac` +0.0592, 2.45 SE — the column the fix touches most), 6 moved more than 1 SE, and the L2 norm of the whole weight change is **0.216**. The joint file moved less: largest term `terrainSetupHelpsPartner` +0.102, L2 **0.128**.
- 3. The combined change is indistinguishable from zero on held-out human-click prediction.** C – A contains zero on both. The fix was worth making because it is a fact about the game that the feature function was getting wrong on 10.72% of turn-boards — that is the whole justification and it does not need a metric to support it.

What this does NOT say. Top-1 agreement with a human click is not a win rate. Nothing here measures whether MILTANK plays better; that is an H2H and it belongs to SEARCH. And the leaf is a separate model from MAG — §1's finding that the leaf is worse than a coin is untouched by any of this.

13a. THE FIT IS NOT RUN IN THE ENVIRONMENT THE BOT PLAYS IN, and it is 20x the weather defect

This is the headline of the refit, not a caveat on it. CLAUDE.md's rule is *fitting environment and playing environment must match*, recorded because MAG's weights were once fitted with the sheet visible while the bot played without it. **The mismatch is back, pointing the other way, and nothing was watching for that direction.**

```
engine/fit_policy.js:376   board.setSheet(side, m.species, { nature, item })
engine/magnemite.js:522   this.board.setSheet(m[1], sp, { nature, item, ability, moves })
```

`Board.switchIn` copies all four onto the active mon, and `dmgMon`, `effAbility` and `movePriority` read them. So the LIVE player sees a sharper board than the fit ever did: the fit prices every opponent on the dataset's representative moveset and on Smogon's per-species ability odds, while an open-sheet game hands the player the declared four moves and the declared ability.

The sheets carry it. 14,400 sheet entries over 1,200 corpus games: 100.0% declare an ability, 100.0% declare four moves. This is not information the fit lacks — it is information the fit is handed and drops.

Measured the same way §11 was, one process holding both builds of `fit_policy` , identical games and identical decisions:

	weather defect (§11)	the sheet-channel gap
candidate vectors that move	1,768 (0.75%)	37,460 (15.95%)
decisions that move	892 (2.78%)	16,177 of 32,054 (50.47%)
columns that move	14 of 58	20 of 58
games containing a moved vector	238 (19.83%)	1,197 of 1,200 (99.75%)

`switchDiesFirst` (10,013), `diesBeforeMoving` (9,466), `switchSurvives1` (6,632), `dmgFrac` (4,188), `killsThreat` , `switchKOSlow` , `switchSurvives2` , `switchKOFast` , `protectThreatened` , `priority` (1,958 — the declared ability reaching `movePriority`) , `screenValue` , `movesFirst` , `koTarget` , `benchRisk` , `killIsRoll` , `koFirst` , `clickCost` , `passTurnAccrues` , `switchFaster` , `deadNoLastMove` . The choice set is unchanged — the row counts match game for game — so this is purely what the board KNOWS, not what it offers.

It is NOT landed and no second refit was started. Landing it is a one-line change to `fit_policy.js:376` plus a full refit of both files, and it would invalidate the refit reported above on the day it was published. It also needs a question answered first that this measurement does not answer: the fit's decisions come from games where the sheet was public, but MAG must also play the ~half of ladder games where the opponent declines OTS, and a model fitted on four channels degrades differently from one fitted on two when a channel goes missing. That is the Focus Sash lesson — *replacing a hedge with a certainty is only an improvement if you also track what invalidates the certainty* — and it is a decision, not a refresh.

Filed with its size stated, which is the part that was missing: **half of every decision the fit trains on is priced against a board the player does not see.**

13b. THE JOINT LAYER IS REFITTED ON FOUR CHANNELS, AND THE CHANNELS ARE WORTH A LIKELIHOOD GAIN, NOT AN ACCURACY GAIN — 2026-08-05

The other half of §13a's debt, run under Will's go. Three results, each with its instrument named.

The joint refit (`engine/fit_joint.js` , four-channel `joint_rows.js`): 8,856 clean open-sheet games, 101,459 joint turns → 95,886 usable, 77,975 train / 17,911 held out by game, `lambda 0` on held-out. The artifact now carries a `fitEnvironment` block and it says `matches_player: true` by measurement: the declared ability and moves reached the board on **202,343 of 202,918 scored slots (99.7%)** and **395,130 of 396,288 live foe actives (99.7%)**. Held out, predicting the pair: separate decisions `logL` -3.3294 / top-1 10.3%, refit with joint terms zeroed -3.3199 / 9.8%, with the joint terms **-3.2308 / 12.2%**. The chosen pair fell outside the top-6 menu on 11.1% of kept turns. `feature_fixture --check` passes on the new artifact. No before/after against the presheet joint vector is quoted because none was measured — the presheet run published no held-out table to its artifact, and comparing two logs would be comparing two samples.

What the two extra channels are worth at the marginal layer (`engine/sheet_channel_value.js` , arm A = release `d3d04b669e18` 's two-channel incumbent, 44,982 paired held-out decisions over 1,789 games, 10,000 game-bootstrap resamples). The first run **VOIDED itself** — ENGINE saved `engine/medicham2-browser.js` mid-run and the instrument recorded `void: true` — and the second run is clean, with every deterministic figure identical between the two:

paired difference	logL/decision	top-1 points
B – A the information alone, weights frozen	+0.002853 [0.001611, 0.004072]	+0.009 [-0.140, +0.157]
C – B the refit, given the information	+0.002234 [0.001638, 0.002831]	+0.165 [0.029, 0.299]
C – A everything vs what shipped	+0.005087 [0.003854, 0.006331]	+0.173 [-0.011, +0.360]

Split-half noise floor of the shipping arm, 20 cuts: **median 0.331 top-1 points** (range 0.012–1.385; the earlier refit's floor was 0.192 on a smaller paired set). Read plainly: **the sheet channels buy a real likelihood gain — every logL interval clears zero — and no demonstrable top-1 gain.** The one top-1 interval that clears zero ($C - B$, +0.165) is half its own noise floor and resolves only because the comparison is paired; the total effect against what shipped contains zero. Same shape as §13: correctness and information first, metric second, and the honest metric statement is "better calibrated per decision, not measurably more often right on the argmax."

The degradation budget did not move, and cannot move by this lever. `fit_joint.turnsDropped` is **5.4929% (5,573 of 101,459) against a 5.49% ceiling — still red.** The dropped turns are unmatched clicks (5,555) and ambiguous mirrors (18); the chosen pair is kept regardless of its rank, so the four-channel w_1 changes which ALTERNATIVES are on the menu, never which turns are kept. The rate crept from 5.4811% when the ceiling was ratcheted (86,242 turns) because the newly ingested games unmatch at 5.56%. The ceiling is untouched; the call on it is Will's.

PORY family regenerated, and `tests/test-site-data-fresh.js` **is GREEN (7/7)** — §5d addendum has the pory-eval numbers; `data/nmf-roles.json` moved 13,258 → 14,808 team-docs, 258 → 263 moves, recon-err 0.8346 → 0.8356, rank 10 unchanged, and both site bundles (`data/pory.js`, `data/nmf.js`) were rewritten by their own generators in the same runs. `data/pory-nn.json` retrained at 6,289 games / 106,782 states (was 6,008 / 102,296): every arm ordering holds — N6 0.6201, NR 0.6132 and LR 0.6064 all still beat the two-feature bar at 0.6229, nonlinearity is still worth ~0.003 and representation ~0.016 — and no living doc quotes these figures (§5c). The first retrain immediately re-red the drift check at 15.1%, the §5f false-denominator class to the letter: its population is the raw-logs subset. Both `engine/pory.py` and `engine/pory_nn.py` now declare `population_ceiling` (the artifact's own generator wrote it; the retrain reproduced every arm to the digit under its seeds), which is what turned the check green rather than a threshold being moved.

Doc and site locations quoting superseded PORY figures, for the propagation pass (grep-verified, historical HANDOFF files excluded as history): `docs/ABRA-whitepaper.md:113` (0.6298 [0.6125, 0.6456]), `docs/SUMMARY.md:77` (same + 0.655), `docs/MODELS.md:358/360/364` (0.9943/1.4080, 0.629799/0.629778, +0.000021 [−0.000013, +0.000056], 925, 4,623), and WEB's `web/stadium.html:506,:728` + `app/stadium.html:506,:728` (the `kadabra` data object and its prose), which render every one of those numbers and are flagged, not edited.

Reading a run

```
node engine/sprt.js <file>
```

Cat the shards together first. **Never read an interim SPRT** — 66.7% became 44%, 57.7% became 50%. The bound exists precisely so you do not have to look. SPRT is valid under continuous monitoring because its boundaries were derived for it; a Wilson interval read repeatedly is not the same thing and does not inherit that property.

The unit is the **decisive pair**, not the game. In a paired run a 1-1 split means the team decided it, not the policy.

Reading a stamp

```
node engine/run_stamp.js --show          data/rollout-r3.json
node engine/run_stamp.js --reconstruct data/rollout-cost.json
```

Every gate artifact has a `<name>.meta.json` beside it saying which configuration produced it. `status.js` prints the headline under the gate line, so the absence of a stamp is on the same screen as the number — R1's +2.91 was quoted for a day against a dump that could not say which of two runs four accuracy points apart it was, and nothing was hidden then either. The fact simply lived in a file nobody opened.

Three things to check before quoting any of it:

- `reconstructed: true` means **inferred from a commit, not observed**. Read `confidence`, which publishes the gap in seconds between the artifact's own timestamp and the commit that carried it.
- `git.dirty: true` means the commit id does not describe what ran. Trust `source_digests`.
- `source_digests` hashes **worktree bytes**; `git.blobs` names git objects. On Windows those differ by line-ending translation — `data/engine-data.js` does — so never compare one to the other.

`writeStamp()` is the only mode worth trusting, because only the run knows its own settings. `reconstruct()` exists for the artifacts that predate it and labels itself on every line.

Running the backtest

```
SHOWDOWN_PATH=C:/Users/willj/Projects/Pokemon/pokemon-showdown node engine/backtest_winrate.js
```

About 13-18 minutes, one process, on 6,886 clean games (809s and 1,046s on two runs). `MAXG=200` thins it for a smoke run, and the artifact records the n it actually scored, so a thinned run cannot be mistaken for a published one. It writes `data/winrate-backtest.json` and the per-game rows beside it. Every seeded configuration reproduces bit-identically across runs; the unseeded legacy `winProb2` arm moved 0.26 accuracy points between two full runs, which is its run-to-run floor.

It stamps the sha256 of every source the leaf reads. `status.js` re-hashes those and prints `CURRENT` or `PRE-CHANGE`, which is a comparison rather than an mtime inference — a checkout moves an mtime without moving code, and the 2026-08-02 artifact was quoted for two days against an engine that had gained "one mega per side" in between.

Done looks like

- `status.js` prints a leaf calibration line that is fresh, adequately powered, and states the reliability curve rather than a verdict string. **Done 2026-08-04** — and the answer is that the leaf is worse than a coin.
- `provenance.js --strict` exits zero.
- Every gate R1–R4 has an artifact. **Done 2026-08-04** — and R1's turned out to disagree with the prose it replaced. An artifact per gate is the floor, not the goal.
- Every gate artifact says which configuration produced it. **Done 2026-08-04 for R2 and R3** via `engine/run_stamp.js`; R1's dump has its own inline copy of the same shape and should call the module. An artifact that records its build still is not enough on its own: R3 records its build and **not its control**, and a divergence rate without the self-disagreement floor beside it is a headline, not a result.
- `REFIT OWED` is either clear or has a dated reason next to it.